POOJA KIRAN

Security Engineer | Application Security | AI & Agent Security | Cloud IAM

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Professional Summary

Security engineer with an M.S. in Information Technology who identifies security flaws, fixes them, and tests the results. Reviewed **14 security repositories**, documented **1 P0 and 5 P1 findings** in a dated source/CI audit, and tested an **MCP security gateway** with **782 passing tests** and **81.93% statement coverage** in a cited Python 3.12 CI run.

Professional & Applied Experience

Independent AI Security Researcher & Engineer Aug 2024 – Present

Self-Directed Security Engineering | Tempe, AZ

- **Reviewed** source code, tests, and **CI workflows** across 14 owned security repositories; recorded **1 P0 and 5 P1 findings**, fixes, and residual risks in a September 2026 security audit.
- **Fixed** unsafe **MCP gateway authorization** responses so explicit 4xx denials block tool execution; added regression checks for client authentication and failure handling.
- **Tested** security changes with **pytest**, **CodeQL**, dependency scans, and **GitHub Actions**; separated passing test evidence from unresolved design and deployment limits.

Business & Compliance Lead | AEROSEC Aug 2025 – Dec 2025

Honeywell Aerospace × ASU | Technology Innovation Lab (TIL) Externship | Tempe, AZ

- **Interviewed** 4 of 12 industry participants over **Zoom**, including a Microsoft cybersecurity specialist, former airline ticketing professional, founder, and former propulsion engineer; documented vendor-access, audit-record, and service-disruption risks.
- **Assigned** tasks to three peer project leads working from separate homes; reviewed technical research, market research, and prototype work during online meetings and reported progress to mentors.
- **Defined API access**, encryption, **audit logging**, and vendor-risk requirements for airline connections using interview findings; drafted a **2–4-week pilot** with owners and success measures.
- **Compared** four security platforms, identified **three target customer segments**, and wrote a **proposed $120,000 first-year budget** with a **12-month cost plan**.

Graduate Teaching Assistant | IT Grader Jan 2025 – Oct 2025

Ira A. Fulton Schools of Engineering | Arizona State University | Tempe, AZ

- **Graded** approximately **85 undergraduate web-programming submissions per semester** and **52 graduate cybersecurity assignments** using instructor rubrics.
- **Wrote** corrective feedback identifying **secure coding**, **risk assessment**, and **compliance** gaps so students could address the issues against grading criteria.

Selected Security Engineering Projects

MCP Agent Security Gateway | Python, FastAPI, MCP/JSON-RPC, Elastic SIEM Oct 2025 – Oct 2026

- **Built** a default-deny **MCP/JSON-RPC** gateway with capability checks, **anti-SSRF** rules, 69 named prompt-injection signatures, hash-chained audit records, and nine **Elastic Security** detection rules.
- **Verified** agent-to-tool enforcement and failure paths using **Python 3.12 CI**; recorded **782 passing tests** and **81.93% statement coverage** in the cited run.

AWS Agent Identity Guard | Python, AWS IAM, SARIF 2.1.0 Apr 2025 – Oct 2026

- **Developed** 25 **AWS IAM** static-analysis rules for broad grants, **iam:PassRole**, **sts:AssumeRole**, and trust-policy risks; exported findings in **SARIF 2.1.0** for CI review.
- **Validated** the rule suite in cited CI: **244 tests passed** and three credential-dependent tests were skipped; did not claim live AWS effective-permission results.

Selected Security Engineering Projects

HF Model Provenance Scanner | Python, SafeTensors, GGUF, ONNX, Keras Nov 2024 – Oct 2026

- **Built** non-executing checks for unsafe serialization, embedded code, **model provenance**, and dependencies across **SafeTensors**, **GGUF**, **ONNX**, and **Keras** inputs.
- **Detected** suspicious content in **33/33 curated adversarial fixtures**; confirmed **260 passing CI tests** and **75.67% statement coverage** without describing fixtures as real-world detection rates.

Attack Detection Engine / Aegis HealthSec | Python, FastAPI, PostgreSQL, ARM/CAN Aug – Oct 2026

- **Extended MITRE ATT&CK v19** detection with offline **CAN**, **C/C++ firmware**, and **ARM ELF32** analysis using synthetic lab inputs and reproducible tests.
- **Integrated** clinical-access detection and a **FHIR R4 AuditEvent** parser with event normalization, deterministic correlation, and **PostgreSQL** incident persistence on unmerged development branches.
- **Tested** an authenticated **FastAPI** investigation workflow with analyst actions and audit history; Milestone E reported **296 local tests passing** with real PostgreSQL, while hosted CI remained blocked.

Technical Skills

Security reviews and risk: Security design review, **threat modeling**, risk analysis, risk assessment, **API authorization**, web-services security, software testing, CI security checks

Cloud, network, operating systems: AWS IAM, DNS, HTTP, TLS, Zeek/Suricata-style telemetry, Linux /proc TCP, **HMAC-SHA256**, **Ed25519**

Engineering and leadership: Python, **C++**, **FastAPI**, **Docker**, **pytest**, **GitHub Actions**, **CodeQL**, SARIF, **Elastic SIEM**, remote peer coordination, compliance research, budgeting

Video content protection (concepts): DRM license flows, encrypted HLS/DASH, CloudFront signed URLs, AWS MediaPackage/SPEKE

Education

Arizona State University May 2026
M.S., Information Technology (Security) | GPA: **3.87/4.00**

M. S. Ramaiah University of Applied Sciences Aug 2023
B.Tech., Computer Science & Engineering

Certifications & Training

AWS Academy Graduate - Cloud Architecting Apr 2025

AWS Academy Graduate - Cloud Security Foundations Nov 2025